DQH-BAN: DUAL-STREAM QUANTUM HEALTHCARE INTELLIGENCE

STREAM A: Quantum Cybersecurity

Network Flows -> Multi-Scale 1D-CNN
-> TCN -> BiGRU -> Attention
-> Angle Encoding -> 4-Qubit VQC
-> Residual Fusion -> Dual Heads

STREAM B: Biomedical Intelligence

Physiological Signals (ECG, PPG, Resp)
-> Bandpass Filtering & SQI Gating
-> 1D-CNN + GRU Encoders
-> Modality-Masked Gated Fusion
-> AF / Non-AF Classification

HYPERLEDGER FABRIC & SECURITY ORCHESTRATION

Device Auth & Access Control | AES-256-GCM Encrypted Off-Chain Vault | SHA-256 Anchor Commitments